

- 6 Nuclei of an isotope of samarium (Sm) each contain 62 protons and 85 neutrons.

(a) State the nuclide notation in the form ${}^A_Z\text{X}$ for this isotope of samarium.

[1]

- (b) This isotope of samarium is radioactive and decays by emitting particles. Gamma-radiation is **not** emitted. The energy spectrum of the emitted particles is shown in Fig. 6.1.

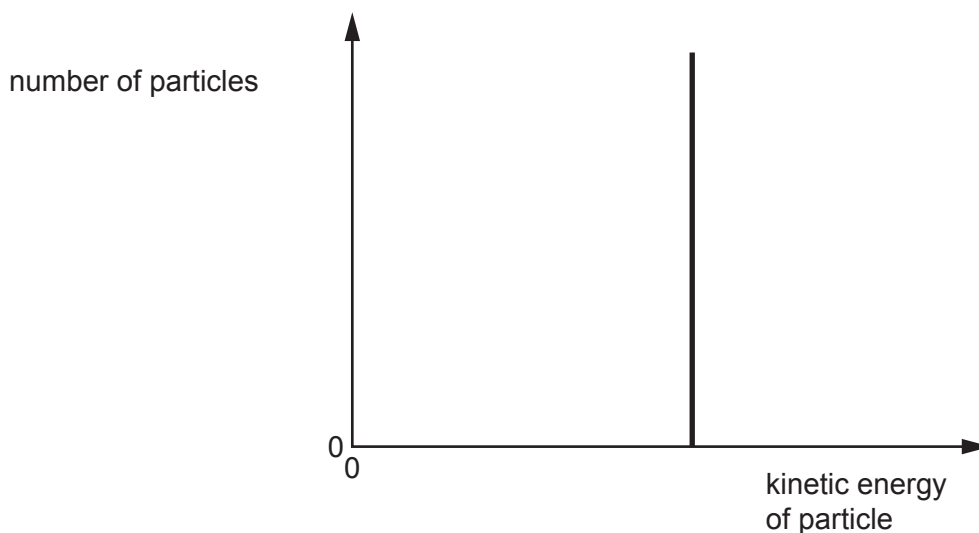


Fig. 6.1

- (i) Explain how Fig. 6.1 shows that this isotope of samarium emits α -particles and does **not** emit β -particles.

.....

 [2]

- (ii) This isotope of samarium decays to an isotope of neodymium (Nd).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[2]

- (c) A baryon is composed of three quarks which all have different flavours. The baryon has a charge of 0.

Two of the quarks in the baryon are an up quark and a bottom quark.

- (i) Determine, in terms of the elementary charge e , the charge on the third quark in the baryon.

charge = e [2]

- (ii) State a possible flavour for the third quark in the baryon.

..... [1]

[Total: 8]